**The due date for submitting this assignment has passed.**
**Due on 2026-04-01, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

You have last submitted on: 2026-04-01, 07:55 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

1) Consider the vector spaces $V$ and functions $\langle .,. \rangle : V \times V \rightarrow \mathrm{R}$ defined as follows: ***1 point***

i) $V = \mathrm{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$.
ii) $V = M_{2\times 2}(\mathrm{R})$ and $\langle A, B \rangle = Tr(AB)$,
where $Tr(M)$ denotes the trace of a matrix $M$, i.e., the sum of the diagonal elements of $M$.
iii) $V = M_{2\times 1}(\mathrm{R})$ and $\langle A, B \rangle = Tr(AB^t)$,
where $Tr(X)$ denotes the trace of a matrix $X$, i.e., the sum of the diagonal elements of $X$. $Y^t$ denotes the transpose of matrix $Y$.
iv) $V = \mathrm{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_2 + x_2y_1$.

- [x] (i) is an inner product.
- [x] (ii) is an inner product.
- [ ] (iii) is an inner product.
- [ ] (iv) is an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(i) is an inner product.
(iii) is an inner product.

2) Consider two linear transformations $T$ and $S$ from $\mathrm{R}^2 \rightarrow \mathrm{R}^2$ defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$. Let $A$ and $B$ be matrix representations of linear transformations $T$ and $S$ with respect to the standard bases of $\mathrm{R}^2$ respectively. Consider the following statements: ***1 point***

**P:** If $c = 1$, then $A$ and $B$ are similar matrices.
**Q:** If $c = 2$, then $A$ and $B$ are similar matrices.
**R:** If $c = 1$ and $P^{-1}AP = B$, then $P$ can be the matrix $\begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$.
**S:** If $c = 1$ and $P^{-1}AP = B$, then $P$ can be the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.
**T:** If $c = 1$, then there are infinitely many $P$ satisfying the equation $P^{-1}AP = B$.

Which of the following options are true?

- [x] $P$ is true but $Q$ is false.

- [ ] Both $P$ and $Q$ are true.
- [ ] Both $R$ and $S$ are true.
- [x] $R$ is false but $S$ is true.
- [x] $T$ is true.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$P$ is true but $Q$ is false.
$R$ is false but $S$ is true.
$T$ is true.

3) Let $\langle ., . \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$. Which one of the following options is (are) true for the vector $\gamma \in \mathbb{R}^2$, such that $\langle \alpha, \gamma \rangle = 4$ and $\langle \beta, \gamma \rangle = 8$, where $\alpha = (3, 1)$ and $\beta = (6, 2)$. ***1 point***

- [ ] No such $\gamma$ exists.
- [x] There are infinitely many such vectors which satisfy the properties of $\gamma$.
- [ ] $\gamma$ is unique in $\mathbb{R}^2$.
- [x] Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of $\gamma$.
- [ ] $(1, 1)$ is the only vector which satisfies the properties of $\gamma$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

There are infinitely many such vectors which satisfy the properties of $\gamma$.
Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of $\gamma$.

4) Let $\langle ., . \rangle$ denote the standard inner product on $\mathbb{R}^2$ , i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ and let $v \in \mathbb{R}^2$. Consider a linear transformation $T_v : \mathbb{R}^2 \to \mathbb{R}$ defined as: ***1 point***

$$T_v(u) = \langle u, v \rangle,$$

where $v \in \mathrm{R}^2$. Which of the following options is (are) true for $T_v$?

- [ ] $T_v$ is one-one for all $v = 0 \in \mathrm{R}^2$.
- [x] $T_v$ is onto for all $v = 0 \in \mathrm{R}^2$.
- [ ] $T_v$ is onto for all $v \in \mathrm{R}^2$.
- [x] $T_v$ is not one-one for every $v \in \mathrm{R}^2$.
- [x] There exists a $v \in \mathrm{R}^2$ such that $T_v$ is not onto.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$T_v$ is onto for all $v = 0 \in \mathrm{R}^2$.
$T_v$ is not one-one for every $v \in \mathrm{R}^2$.
There exists a $v \in \mathrm{R}^2$ such that $T_v$ is not onto.

5) Let A and B be square matrices of the same order $n$. Which of the following statements are sufficient to conclude that A is equivalent to B. ***1 point***

- [ ] det(A) = det(B).
- [x] Nullity(A) = Nullity(B).
- [ ] The set of columns of $A$, and the set of columns of $B$ are both linearly dependent subsets of $\mathrm{R}^n$.
- [x] There exists a matrix $C$ such that $A$ is equivalent to $C$, and $B$ is equivalent to $C$.
- [x] There exist invertible matrices $P$ and $Q$ of order $n$ such that $PA = BQ$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Nullity(A) = Nullity(B).
There exists a matrix $C$ such that $A$ is equivalent to $C$, and $B$ is equivalent to $C$.
There exist invertible matrices $P$ and $Q$ of order $n$ such that $PA = BQ$.

6) Let $L$ and $L'$ be affine subspaces of $\mathrm{R}^3$, where $L = (0,1,1) + U$ and $L' = (0,1,0) + U'$, for some vector subspaces $U$ and $U'$ of $\mathrm{R}^3$. Let a basis for $U$ be given by $\{(1,1,0),(1,0,1)\}$ and a basis for $U'$ be given by $\{(1,0,0)\}$. Suppose there is a linear transformation $T : U \to U'$ such that $(1,0,1) \in \ker(T)$ and $T(1,1,0) = (1,0,0)$. An affine mapping $f : L \to L'$ is obtained by defining $f((0,1,1) + u) = (0,1,0) + T(u)$, for all $u \in U$. Which of the following options are true? ***1 point***

- [x] $L = \{(x, y+1, x-y+1) \mid x, y \in \mathrm{R}\}$.
- [ ] $L' = \{(x, y+1, 0) \mid x, y \in \mathrm{R}\}$.
- [ ] $L = \{(x-y, y+1, x-y+1) \mid x, y \in \mathrm{R}\}$.
- [ ] $L = \{(x, x+1, y+1) \mid x, y \in \mathrm{R}\}$.
- [x] $f(x, y+1, x-y+1) = (y, 1, 0)$
- [ ] $f(x-y, y+1, x-y+1) = (x, y+1, 0)$
- [ ] $f(x, x+1, y+1) = (y, 1, 0)$
- [ ] $f(x, y+1, x-y+1) = (0, 1, y)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$L = \{(x, y+1, x-y+1) \mid x, y \in \mathrm{R}\}$.
$f(x, y+1, x-y+1) = (y, 1, 0)$

7) Let $\theta$ be the angle between the vectors $u = (4,7,3)$ and $v = (1,2,-6)$, then what will be the value of $cos(\theta)$?

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

***1 point***

8) Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\}$$

of $\mathrm{R}^3$ with usual inner product. Suppose $v = (x, y, \frac{3x+y}{5}) \in V$ is written as $v = c_1v_1 + c_2v_2 + c_3v_3$, such that $c_1 + c_2 = 4$. What will be the value of $c_3$?

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

***1 point***

9) Let $V = \mathrm{R}^2$ be a vector space. Consider two inner products $\langle ., . \rangle_1$ and $\langle ., . \rangle_2$ on $V$ defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1x_2 - x_1y_2 - x_2y_1 + 4y_1y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1x_2 + 2y_1y_2.$$

If $\langle (a, b), (4, 3) \rangle_1 = 35$ and $\langle (a, b), (4, 3) \rangle_2 = 60$, then find the value $a + 2b$.

11

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 11

***1 point***

10) Let $V = \mathrm{R}^3$ be the inner product space with usual inner product. If $\theta$ is the angle between $(15, 8, 18)$ and $(a, b, c)$ where $15a + 8b + 18c = 0$ and $(a^2 + b^2 + c^2) = 0$, then find the value of $\cos\theta$.

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

***1 point***

11) Consider a vector $v = (x - 20, 2, 1) \in \mathrm{R}^3$. Find the value of $x$ so that the length of the vector $v$ is minimum.

20

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

***1 point***

12) Consider the following subsets of $\mathrm{R}^3$, with $\langle\ ,\ \rangle$ used to denote the standard dot product on $\mathrm{R}^3$, and $\|\ \|$ used to denote the standard length function.

$S_1 = \{v \in \mathrm{R}^3 \mid \langle v, (0,0,1)\rangle = 2,\ \|v\| = 1\}$

$S_2 = \{u \in \mathrm{R}^3 \mid u \in \mathrm{Span}(\{(1,1,1)\}),\ \|u\| = 1\}$

If $n_1$ and $n_2$ denote the number of elements in $S_1$ and $S_2$, respectively, then find the value of $n_1 - n_2$.

-2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -2

***1 point***